Logistic Regression with Regularization

1) Why Regularization?
Regularization prevents overfitting by keeping weights small.
It adds a penalty term to the cost function.

2) Model Formula:
f(x) = sigmoid(w.x + b)

3) Regularized Cost Function:
J = Log Loss + (lambda / 2m) * sum(w^2)
Only weights (w) are regularized, NOT bias (b).

4) Regularized Gradient:
dj_dw = (1/m) * sum((f - y) * x) + (lambda/m) * w
dj_db = (1/m) * sum(f - y)

5) Key Coding Logic:

Regularized Cost:
reg_cost = (lambda_ / (2*m)) * np.sum(w**2)
total_cost = cost + reg_cost

Regularized Gradient:
dj_dw = dj_dw + (lambda_/m) * w

6) Gradient Descent Update:
w = w - alpha * dj_dw
b = b - alpha * dj_db

7) Intuition:
- Normal gradient fits the data
- Regularization pulls weights toward 0
- Large lambda → simpler model
- Small lambda → more flexible model

Final Understanding:
Regularization balances fitting the data and keeping weights small.